

HDOC DAY 29

AIM :

To write a simple menu-driven C program to count even, odd, positive, negative, and zero elements in an array, search for an element using linear search, merge two arrays, and exit the program based on user choice.

ALGORITHM

1. Start

2. Display Menu Options:

- a. Option 1: Count even, odd, positive, negative, and zeros in an array.
- b. Option 2: Search for an element using Linear Search.
- c. Option 3: Merge two arrays.
- d. Option 4: Exit.

3. Take Choice: Read the user's choice using a switch-case inside a while loop.

4. If Choice = 1:

- a. Input the size of the array and the array elements.
- b. Initialize counters: even = 0, odd = 0, pos = 0, neg = 0, zero = 0.
- c. Loop through the array elements:
 - i. Check if element > 0 (increment pos), < 0 (increment neg), or == 0 (increment zero).
 - ii. Check if element % 2 == 0 (increment even), else (increment odd).
- d. Print all counter values.

5. If Choice = 2:

- a. Input the size of the array and the array elements.
- b. Input the number to search (key).
- c. Loop through the array from start to end.
- d. If array element equals key, set found = 1 and store index.
- e. If found == 1, print success message and position, else print "Element not found".

6. If Choice = 3:

- a. Input size and elements of First Array.
- b. Input size and elements of Second Array.
- c. Copy all elements of the first array into a Third Array.
- d. Copy all elements of the second array into the Third Array after the first array's elements.
- e. Print the merged Third Array.

7. If Choice = 4:

- a. Stop execution / Exit loop.

8. Default Case:

- a. Print "Invalid choice! Try again."

9. End

C code :

```
#include <stdio.h> /
int main() {
    int choice;

    while(1) {
        printf("\n=== MENU ===\n");
        printf("1. Count Even, Odd, Positive, Negative, and Zero elements\n");
        printf("2. Search element using Linear Search\n");
        printf("3. Merge two arrays\n");
        printf("4. Exit\n");
        printf("Enter your choice (1-4): ");
        scanf("%d", &choice);

        switch(choice) {
            case 1: {
                int n, i;
                int even = 0, odd = 0, pos = 0, neg = 0, zero = 0;

                printf("Enter number of elements in array: ");
                scanf("%d", &n);

                int arr[n];
                printf("Enter %d elements:\n", n);
                for(i = 0; i < n; i++) {
                    scanf("%d", &arr[i]);
                }

                for(i = 0; i < n; i++) {
                    // Check positive, negative, zero
```

```

        if(arr[i] > 0) {
            pos++;
        } else if(arr[i] < 0) {
            neg++;
        } else {
            zero++;
        }

        // Check even, odd
        if(arr[i] % 2 == 0) {
            even++;
        } else {
            odd++;
        }
    }

    printf("\n--- Results ---\n");
    printf("Positive numbers: %d\n", pos);
    printf("Negative numbers: %d\n", neg);
    printf("Zero elements: %d\n", zero);
    printf("Even numbers: %d\n", even);
    printf("Odd numbers: %d\n", odd);
    break;
}

case 2: {
    int n, i, key, found = 0;

    printf("Enter number of elements in array: ");
    scanf("%d", &n);

    int arr[n];
    printf("Enter %d elements:\n", n);
    for(i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    printf("Enter element to search: ");

```

```

scanf("%d", &key);

for(i = 0; i < n; i++) {
    if(arr[i] == key) {
        printf("Element %d found at index %d (Position %d)\n", key, i, i + 1);
        found = 1;
        break;
    }
}

if(found == 0) {
    printf("Element %d not found in array.\n", key);
}
break;
}

case 3: {
    int n1, n2, i;

    printf("Enter size of first array: ");
    scanf("%d", &n1);
    int arr1[n1];
    printf("Enter %d elements for first array:\n", n1);
    for(i = 0; i < n1; i++) {
        scanf("%d", &arr1[i]);
    }

    printf("Enter size of second array: ");
    scanf("%d", &n2);
    int arr2[n2];
    printf("Enter %d elements for second array:\n", n2);
    for(i = 0; i < n2; i++) {
        scanf("%d", &arr2[i]);
    }

    int merged[n1 + n2];

    // Copy first array

```

```
    for(i = 0; i < n1; i++){
        merged[i] = arr1[i];
    }

    // Copy second array
    for(i = 0; i < n2; i++){
        merged[n1 + i] = arr2[i];
    }

    printf("\nMerged Array elements are:\n");
    for(i = 0; i < n1 + n2; i++){
        printf("%d ", merged[i]);
    }
    printf("\n");
    break;
}

case 4:
    printf("Exiting program. Thank you!\n");
    return 0;

default:
    printf("Invalid choice! Please enter a number between 1 and 4.\n");
}
}

return 0;
}
```

CODE SCREENSHOT -

```
#include <stdio.h>

int main() {
    int choice;

    while (1) {
        printf("\n1. Count Elements\n");
        printf("2. Linear Search\n");
        printf("3. Merge Two Arrays\n");
        printf("4. Exit\n");
        printf("Enter choice: ");
        scanf("%d", &choice);

        if (choice == 1) {
            int n, i;
            int even = 0, odd = 0, pos = 0, neg = 0, zero = 0;

            printf("Enter array size: ");
            scanf("%d", &n);

            int a[100];
            printf("Enter %d numbers: ", n);
            for (i = 0; i < n; i++) {
                scanf("%d", &a[i]);
            }

            for (i = 0; i < n; i++) {
                if (a[i] > 0)
                    pos++;
                else if (a[i] < 0)
                    neg++;
                else
                    zero++;

                if (a[i] % 2 == 0)
                    even++;
                else
                    odd++;
            }

            printf("Positive: %d\n", pos);
            printf("Negative: %d\n", neg);
            printf("Zero: %d\n", zero);
            printf("Even: %d\n", even);
            printf("Odd: %d\n", odd);
        }
        else if (choice == 2) {
            int n, i, key, found = 0;

            printf("Enter array size: ");
            scanf("%d", &n);

            int a[100];
            printf("Enter %d numbers: ", n);
            for (i = 0; i < n; i++) {
```

□

```
        for (i = 0; i < n; i++) {
            scanf("%d", &a[i]);
        }

        printf("Enter number to search: ");
        scanf("%d", &key);

        for (i = 0; i < n; i++) {
            if (a[i] == key) {
                printf("Found at position %d\n", i + 1);
                found = 1;
                break;
            }
        }

        if (found == 0) {
            printf("Not found\n");
        }
    }
    else if (choice == 3) {
        int n1, n2, i, c[200];

        printf("Enter size of first array: ");
        scanf("%d", &n1);
        int a[100];
        printf("Enter elements: ");
        for (i = 0; i < n1; i++) {
            scanf("%d", &a[i]);
```

```

        scanf("%d", &a[i]);
    }

    printf("Enter size of second array: ");
    scanf("%d", &n2);
    int b[100];
    printf("Enter elements: ");
    for (i = 0; i < n2; i++) {
        scanf("%d", &b[i]);
    }

    for (i = 0; i < n1; i++) {
        c[i] = a[i];
    }

    for (i = 0; i < n2; i++) {
        c[n1 + i] = b[i];
    }

    printf("Merged array: ");
    for (i = 0; i < n1 + n2; i++) {
        printf("%d ", c[i]);
    }
    printf("\n");
}
else if (choice == 4) {
    printf("Exiting...\n");
    break;
}
else {

```

```

    }
    else {
        printf("Invalid choice!\n");
    }
}

return 0;
}

```

OUTPUT SCREENSHOT :

```
[manalithakar@Manalis-MacBook-Air ~ % nano 29q.c
[manalithakar@Manalis-MacBook-Air ~ % gcc 29q.c -o 29q
[manalithakar@Manalis-MacBook-Air ~ % ./29q
```

```
1. Count Elements
2. Linear Search
3. Merge Two Arrays
4. Exit
Enter choice: 1
Enter array size: 5
Enter 5 numbers: 12 -5 0 8 -3
Positive: 2
Negative: 2
Zero: 1
Even: 3
Odd: 2
```

```
1. Count Elements
2. Linear Search
3. Merge Two Arrays
4. Exit
Enter choice: 2
Enter array size: 4
Enter 4 numbers: 10 20 30 40
Enter number to search: 30
Found at position 3
```

```
1. Count Elements
2. Linear Search
3. Merge Two Arrays
4. Exit
Enter choice: 3
Enter size of first array: 3
Enter elements: 5 10 15
Enter size of second array: 2
Enter elements: 20 25
Merged array: 5 10 15 20 25
```

```
1. Count Elements
2. Linear Search
3. Merge Two Arrays
4. Exit
Enter choice: 4
Exiting...
```

```
manalithakar@Manalis-MacBook-Air ~ % █
```